Coaching How To Search

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Abstract: Searching is commonly addressed by either directly programming a machine to do so or letting it explore the search state space and craft a path "on its own", as in Reinforcement Learning. Both approaches place most of the computational and / or cognitive load to one end, be it the programmer or the machine learner. An alternative way, balancing load between the two learning ends, is to have a human or machine coach possesing explicit knowledge on the underlying search problem to provide adequate advice upon a machine learner's request for further knowledge. In this paper, building on previous work on Machine Coaching, specialized in the particular context of searching, a novel learning paradigm is presented, investigating its efficacy and conformity. Potential next steps and real-world applications of the proposed paradigm are also discussed.

1 Introduction

Artificial Intelligence (AI), and Generative AI (GenAI) in particular, has found plentiful applications in nearly all aspects of everyday life. On several occasions, the underlying tasks can be formulated as search problems, in the sense of having an AI agent move around a state space, starting from an initial state and crafting a path towards a goal state. Examples of this might include route finding, puzzle solving, shopping planning, code debugging, clinical counseling and so on. Furthermore, restrictions are put on the shape of the *optimal* path, in accordance to the nature of the task at hand, e.g., seeking a shortest path for route finding (Liu, 1996; Migel et al., 2024), finding a particular kind of solution to a puzzle (Mutalova et al., 2023), or strictly following a therapist's guidance to provide further advice to a client (Sedlakova and Trachsel, 2023; Fiske et al., 2019).

Programming and Reinforcement Learning (RL) are two common ways enabling a machine to solve search problems. Both can be described in the context of an expert, the programmer and the reward function, respectively, interacting with a learner, instilling knowledge to it. Considering a human or machine coach possessing explicit knowledge on a search problem in the expert's role, and a learner capable of receiving advice on its actions provides a learning framework where expert cognitive effort is reduced compared to directly programming the learner's behavior, since in this case the learner undertakes part of this task. Similarly, receiving explicit expert feedback lifts computational load from the learner compared to pure RL. Hence, a coaching based learning setting might provide a more balanced alternative regarding both expert and learner effort put on crafting an optimal path for a search problem.

Notably, in all such cases, path optimality can also be understood in the context of conformance to the search policy of a coach, be it explicit, e.g., in a therapeutic setting, or implicit, e.g., in the case of route finding. This kind of formulation allows one to also control, under appropriate learning semantics, not only *what* the learner learns to accomplish, but also *how* this is accomplished, which might be of higher significance than the reaching the goal itself in certain settings. In this work, such a search coaching paradigm is put forward, inspired by previous work (Michael, 2019; Markos and Michael, 2022), and tested in a controlled artificial setting, shedding more light on the interplay between "what" an agent learns to look for and "how" it actually does so.

2 Related Literature

Knowledge transfer between a machine learner and a human coach has been discussed in various works, such as eXplainable Interactive Learning (XIL), (Teso and Kersting, 2019) which considers a human supervisor in the role of a learning "coach", with advice either correcting mislabeled data, or wrong explanations provided by the learner for otherwise correct predictions. Nevertheless, while in Machine Coaching coach advice can be directly embedded into the learner's knowledge base in an elaboration-tolerant manner (McCarthy, 1998), XIL utilizes an active learning algorithm as a black-box (Settles, 2009) alongside a local post-hoc explainer (Ribeiro et al., 2016). The corresponding explanations are then used to generate additional labeled data for the learner, to help it cure any misunderstandings.

User feedback can also be utilized to proactively reduce the need for data point labeling, as in (Raghavan and Allan, 2007) for Support Vector Machines. In the same spirit, in (Settles, 2011) an interactive framework is proposed, allowing users to label selected data points with the aim of further polishing the training of a document classifier, while in (Zaidan et al., 2007), a machine learning paradigm utilizing user-provided explanations to reduce learning complexity is proposed. Interestingly, in (Zaidan et al., 2007) it is stressed that an alternative to immense data load for performance improvement is *data richness*, i.e., the provision of more meaningful data, e.g., in the form of a correct label accompanied by an explanation for it. *Coactive Learning*, proposed in (Shivaswamy and Joachims, 2015), is another interactive Machine Learning framework where a learner receives user advice that is, inherently, only "slightly better" than the learner's current prediction. The main benefit of the approach, as shown in (Shivaswamy and Joachims, 2015), is that this approach can be integrated into many existing black-box methodologies.

Besides guaranteeing faithfulness and effectively building user trust (Barredo Arrieta et al., 2020), explanation embedding during model training has been reported to improve testing performance (Selvaraju et al., 2019). Moreover, utilizing bilateral explanations, i.e., from the learner towards the coach and vice versa, can further speed up the learning process (Dong et al., 2017), with similar results also obtained in model retraining upon the provision of additional explanations regarding both wrong and correct, but for the wrong reasons, model decisions (Rieger et al., 2019). At last, post-hoc model fine tuning also leads to performance improvements (Ribeiro et al., 2016).

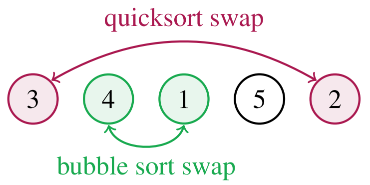
Given the recent spike in the use of Large Language Models (LLMs) in virtually all domains, human-machine interaction methodologies that utilize human expertise to further polish a LLM have been proposed. In (Welz and Lanquillon, 2024) a human knowledge infusion methodology is presented, where expert users provide domain specific knowledge that is used to populate a knowledge base, then utilized by the system in tandem with the LLM to generate responses to user queries. In a similar fashion, in (Wang et al., 2024) a human-LLM collaborative data annotation approach is presented, where the human undertakes the role of the LLM supervisor by revising and correcting the latter's erroneous data labels and / or the corresponding LLM explanations, as in (Teso and Kersting, 2019). In (Kim et al., 2024), human-LLM collaboration is explored in a setting where the user can freely explore, correct, and interact with LLM responses through appropriate programmatic interfaces or interface widgets. A variant of this approach, where human supervisors freely observe and correct LLM annotations in PDF files is also proposed in (Tang et al., 2024). In robotics, human robot teleoperators have been put in coaching roles in hybrid robot-LLM systems, with the aim of system decision verification (Liu et al., 2023). In the context of generative design customization, passive, assistive and proactive human-LLM collaboration schemes have been proposed (Wang et al., 2025), each one for users of different levels of expertise.

It should be noted that knowledge exchange between a coach and a machine learner in the form of context-driven advice giving based on the learner's decisions has been described as soon as in (McCarthy, 1959). Recently, it has been formalized in the context of *Machine Coaching* (Michael, 2017; Michael, 2019) under the Probably Approximately Correct (PAC) model (Valiant, 1984).

While AI system verification and faithful human expertise embedding have been quite researched, it is usually the case that the emphasis is put on what, eventually, the machine learner can achieve and how successful it is. Nevertheless, there are plentiful applications where such a desiteratum comes second to *"how"* the machine learner achieves its goals. Deep neural architectures and LLMs, even if prevalent in most applications due to their efficiency and efficacy, are not necessarily the best candidates to undertake the task of exact human knowledge representation, in part due to the lack of solid formal guarantees on their behavior. On the contrary, argumentation-based approaches (Dung, 1995; Prakken, 2010) come with such guarantees "out-of-the-box", and could provide a useful tool in cognitive agent design in such cases.

3 Background and Preliminaries

Utilizing learning semantics presented in previous work on Machine Coaching (Michael, 2019; Markos and Michael, 2022), its particular application on

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Rbf :: (3,4,1,5,2) implies swap(4,1);
Rbp :: (_,4,1,_,_) implies swap(4,1);
Rqf :: (3,4,1,5,2) implies swap(3,2);
Rqp :: (3,_,_,_,2) implies swap(3,2);
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Figure 1: Examples of full (f) and partial (p) advice from $\mathtt{P}_{\text{bubble}}$ (b) and $\mathtt{P}_{\text{quick}}$ (q) on state $(3,4,1,5,2)$.

search coaching is empirically explored, by having a machine learner iteratively seek advice by a machine coach, emulating a human coach, as in (Markos et al., 2022), with the aim of iteratively improving its search capacity, guided by the coach's explicit search policy. Given a fixed goal state, g, the learner applies its search policy, $\mathtt{P}_{learner}$, to identify a path, p, starting from an arbitrary state, s, and ending at a goal state, g. Upon lack of knowledge, the learner seeks advice from its coach, which responds according to its own policy, $\mathtt{P}_{coach}$, providing a single piece of advice to the learner. Each piece of advice has the form of an "if-then" rule, where the rule's *body* ("if" part) is state-dependent and its *head* (the "then" part) is always an action to be undertaken by the learner. Rules are prioritized by the coach as in previous works (Markos et al., 2022; Markos and Michael, 2022; Ioannou and Michael, 2021) with later received ones by default overriding previous ones in $\mathtt{P}_{learner}$, besides cases where the coach demands otherwise. In this setting, a *coaching step* is a single learner request for advice and the corresponding coach response, while a *coaching session* comprises of the full learning cycle, ending when the learner eventually crafts a path as requested (cf Algorithm 1). In all cases the learner seeks advice upon lack of knowledge for further searching, returning a partial path p starting from s.

Data: s, g, *learner*, *coach*
Result: *learner* with updated search policy;
path from s to g.
do
  $path \leftarrow learner.search\_path(s,g)$;
  $advice \leftarrow coach.evaluate(path,s,g)$;
  $learner.update\_hypothesis(advice)$;
while *advice is not* `null`;

Algorithm 1: Search coaching process. s, g denote the start and goal states, respectively.

The particular case of sorting formulated as a search problem has been considered, implementing

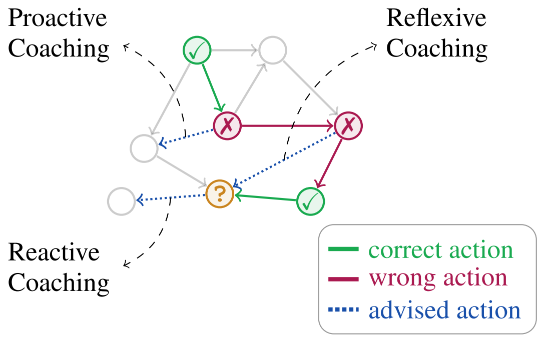
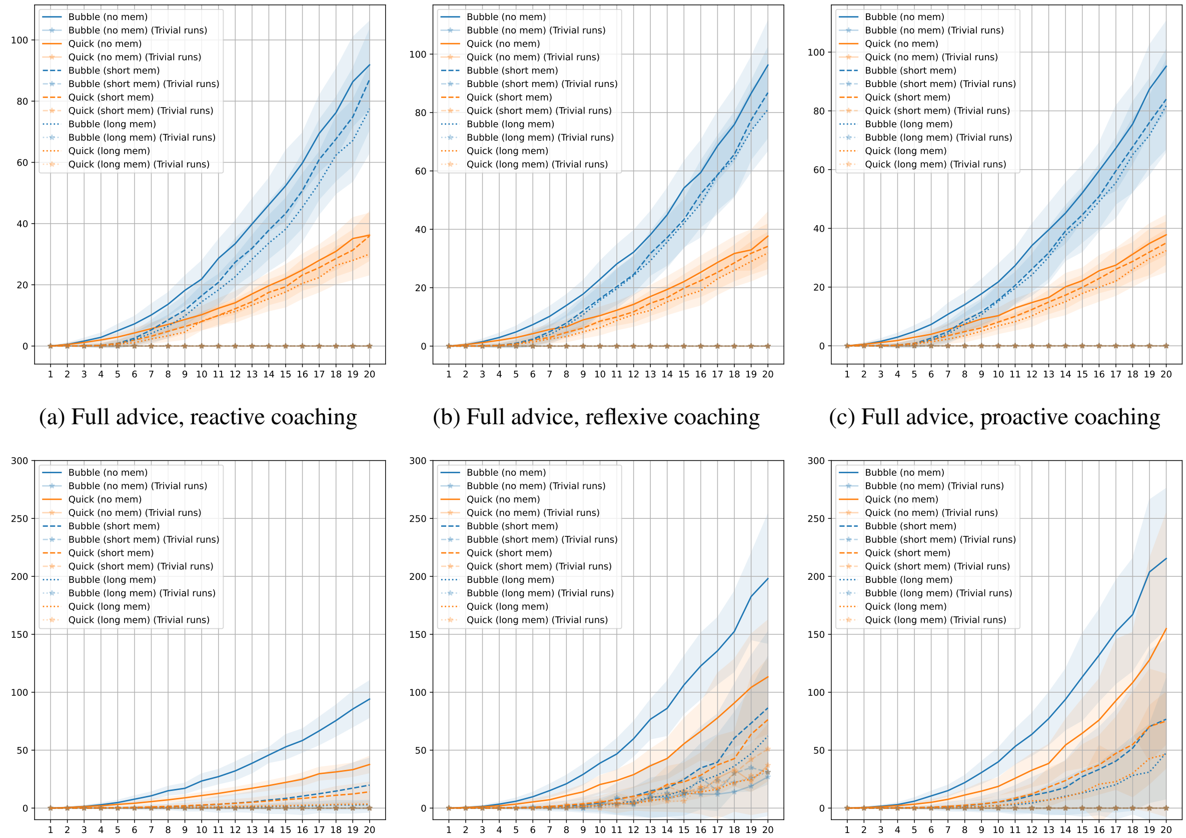


Figure 2: Different coaching styles used in the experimental setup. Upon the learner reaching a state from where there is no knowledge on how to proceed (?), a reactive piece of advice resolves this lack of knowledge, a proactive piece of advice addresses the first wrong decision in the learner's search path, while a reflexive piece of advice addresses a random mistake / lack of knowledge of the learner.

two coaching *policies*, $\mathtt{P}_{\text{bubble}}$, which relies on bubble sort, and $\mathtt{P}_{\text{quick}}$, which relies on quicksort, both resulting to advice corresponding to the next step of the algorithm described by each coaching policy. A state of size n in this case is any permutation of $\{1,2,\ldots,n\}$, with start states being arbitrarily chosen among all possible states of the same size, while the goal state is always $(1,2,\ldots,n)$. Rules in this case have a potentially partial permutation of $\{1,2,\ldots,n\}$ as their body, and a swap action between two items of this permutation as their head, as shown in Figure 1.

In this setting, two advice types are considered: (A1) *full advice*, where each piece of advice depends on the entire state, and; (A2) *partial advice*, where each piece of advice depends on the specific part of the state to be improved. Moreover, for each advice type three learner configurations regarding advice memorization by the learner have been considered: (M1) *no memory* across different iterations for the same state size, n, i.e., coaching sessions and the corresponding learned policies are pairwise independent; (M2) *short memory* across different iterations for the same state size, n, i.e., coaching sessions across different values of n are independent but not within the same value of n, and; (M3) *long memory* across all values of state size, n, i.e., all coaching sessions are correlated, incrementally building on knowledge from all previous test cases. A coach might adopt one of the following *coaching styles*, as illustrated in Figure 2: (C1) *reactive* coaching, where the coach provides advice on the learner's last wrong decision in p; (C2) *reflexive* coaching, where the coach provides advice at a random wrong decision in p, and; (C3) *proactive* coaching, where the coach provides advice on the learner's earliest wrong decision in p.



(a) Full advice, reactive coaching (b) Full advice, reflexive coaching (c) Full advice, proactive coaching

(d) Partial advice, reactive coaching (e) Partial advice, reflexive coaching (f) Partial advice, proactive coaching

Figure 3: Coaching steps as a function of state size, n. Full and partial advice (top and bottom, respectively) with reactive, reflexive and proactive coaching styles are considered (left, middle, and right, respectively), with all possible coach ($\mathtt{P}_{\text{bubble}}$, $\mathtt{P}_{\text{quick}}$) and memory (none, short, long) combinations are shown.

4 Experimental Setup

A series of experiments was run first to verify learning efficacy, where state size, n, ranges from 1 to 20, running $m = 100$ iterations for each value of n. All possible combinations of advice type (full and partial), memory configurations (none, short, long), coaching styles (reactive, reflexive, proactive), and coaching policies ($\mathtt{P}_{\text{bubble}}$ and $\mathtt{P}_{\text{quick}}$) have been considered, resulting into 36 different coaching settings.

In order to explore how much the learner conforms to the coach's instructions in the various settings explored, *coaching conformity* is defined to to be the relative frequency of learner decisions conforming to coach decisions on the same states over all such decisions in a session. First, $m = 20$ repetitions for state size $n \in \{5, 10, 15, 20\}$ are sampled with coaching conformity being computed for each one, across all coach policies, coaching styles, and memory configurations. Then, aiming to shed more light on the parameters affecting coaching conformity, a new type of advice, partial@k, $2 \leq k \leq n$, is introduced, where $k-2$ randomly selected state variables are included into the advice condition, alongside the two required to describe the swap action of interest. Such advice captures intermediate levels of specificity between the two extremes explored so far, with partial@2 corresponding to partial and partial@n to full advice for state size n. Also, *advice coverage* is defined as the percentage of state variables that is included in the condition of a piece of advice. So, e.g., a rule like `R :: (_,2,1,_,5) implies swap(2,5)` has an advice coverage of $3/5 = 60\%$. Again, $m = 20$ iterations for state size $n \in \{5, 10, 15, 20\}$, $2 \leq k \leq n$, for all coaching policies, coaching styles, and memory configurations were run.

5 Results

In Figures 3 the number of coaching steps required to reach the goal state in each case as a function of the state size, n, for all three coaching styles is presented. In all cases, except for partial advice reflexive coaching, the learner completed successfully all coaching sessions with the intended results. Considering reactive coaching and full advice (Fig. 3a), the utilization of either short or long memory does not appear to have any particular effect on the number of coaching steps, which was expected given the restricted applicability of provided advice. On the contrary, partial advice (Fig. 3d) combined with both short and long memory results to fewer coaching steps, which can be symmetrically explained by the broader applicability of the provided advice. Also observe that in the case of full advice, coaching steps are proportional to the big-O complexity of the underlying policy / sorting algorithm, which, again, comes as a result of limited advice applicability, forcing the learner to essentially ask for step-by-step guidance at each separate case. In juxtaposition to the full advice case, utilizing memory significantly reduces the required coaching steps per session, since in this case the learner can effectively apply previously acquired knowledge to new cases. Furthermore, note that in the long memory case (Fig. 3d), coaching steps for both $\mathtt{P}_{\text{bubble}}$ and $\mathtt{P}_{\text{quick}}$ are essentially the same (dotted lines), indicating that in the case of reactive coaching, memorizing all previous advice eventually allows the learner to generalize and autonomously reach the goal state without frequent interaction with its "lazy" supervisor.

Considering proactive coaching, where the learner receives advice on its first wrong decision (Figs. 3c, 3f), observe that full advice imposes the same trends in coaching step growth as with reactive coaching. However, providing partial advice leads to a significantly reduced number of coaching steps per session, as with reactive coaching, since, by utilizing memory, the learner is able to generalize from previous cases. Notably, coaching steps are systematically higher from the corresponding reactive coaching cases (Figs. 3d, 3f). This reflects the more "meticulous" way a proactive coach interacts with the learner, providing (even partial) advice that is more accurate, preventing any unnecessary exploration of the search space, at the cost of interacting more frequently with the learner, compared to a reactive coach, which just reacts to a learner's demand for further knowledge.

Regarding reflexive coaching (Figs. 3b, 3e), the same trends as in proactive coaching are observed. However, in this case, the coach randomly selecting a state to intervene gives rise to the possibility that the same piece of advice might be provided more than once. The number of such sessions is reported as *trivial runs*. As state size grows larger, so does the number of trivial runs, especially in non trivial memory configuration cases, where the learner's policy is typically larger, hence it is, in general, more probable to receive the same piece of advice more than once.

Finally, observe that coaching policy is irrelevant in case the learner is equipped with some kind of memory, regardless of advice type. For full advice this is expected given the restricted advice applicability which prohibits the learner from reusing previous knowledge, while partial advice allows the learner to generalize conforming to $\mathtt{P}_{coach}$, reducing the need for coach feedback. Focusing on partial advice in the absence of memory, reactive coaching results to coaching sessions of size proportional to the coaching policy complexity, as noted above. However, for proactive and reflexive coaching this is not true, as the coach appears to interact with the learner more frequently, indicating that in the reactive case the learner might learn an efficacious but unfaithful policy, which might not occur for other coaching styles.

In Figure 4 average coaching conformity as a function of state size, n, is illustrated. Regardless of memory configuration, full advice combined with reactive coaching result to a virtually fully conformant learner, replicating the coach's policy. Similar results are also obtained for proactive and reactive coaches. However, with partial advice, coaching conformity depends on the coaching style and the corresponding memory configuration. In particular, in the case of reactive coaching, partial advice results to virtually vanishing conformity for all memory configurations. For proactive coaching, however, partial advice results to a constantly high conformity for all examined state sizes, also barely affected by the corresponding memory configuration. Similarly, reflexive coaching also results to higher conformity than reactive coaching, roughly replicating results for proactive coaching. Also, observe how in all cases coach type does not affect conformity, except for partial advice reactive coaching, where any differences appear to vanish as state size grows. Overall, Figure 4 hints towards a trade-off between advice specificity and coaching conformity. The more specific the advice the more conformant the learner to the coaching policy and, conversely, the looser the advice, the more the learner generalizes utilizing previous knowledge.

In Figure 5 coaching conformity against advice coverage for all coaching policies and coaching styles, aggregating over state size, n, is presented. Observe, at first, how coaching policy and memory configuration do not affect coaching conformity,

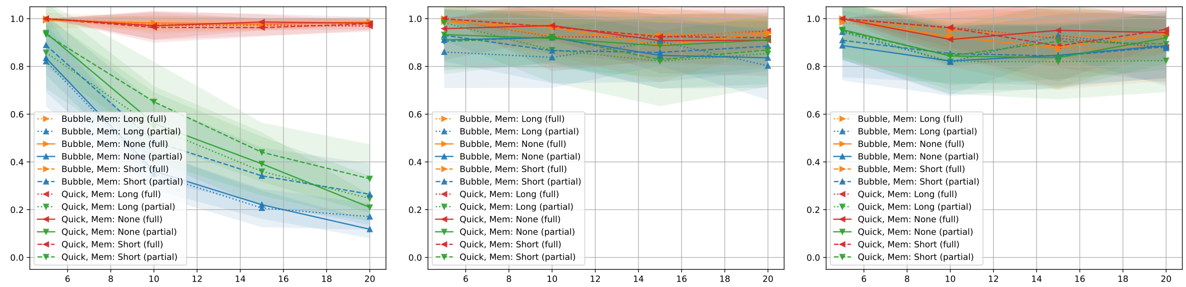


Figure 4: Coaching fidelity against state size for reactive (left), reflexive (middle) and proactive (right) coaching.

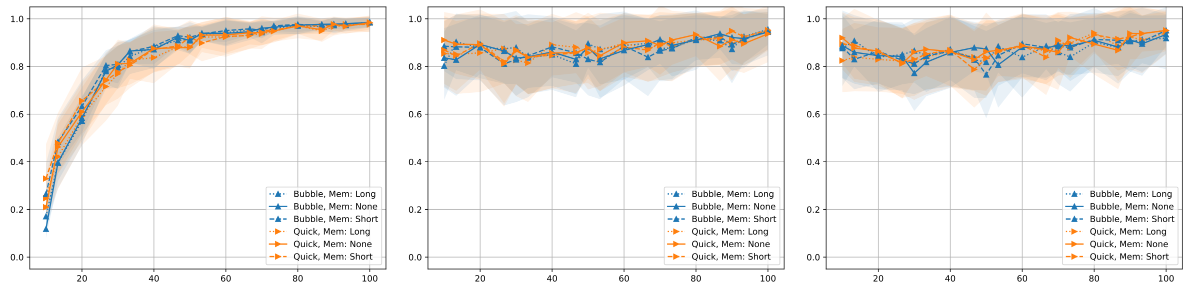


Figure 5: Coaching fidelity against advice coverage for reactive (left), reflexive (middle) and proactive (right) coaching.

while the most significant effect comes from different coaching styles. Indeed, in the case of reactive coaching, lower advice coverage results to the learner eventually reaching the goal state but in a way not conforming to the coach's search policy, while the more specific the provided advice, the more accurate the learner in its searching steps towards the goal state. On the contrary, for proactive and reflexive coaching, coaching conformity is just slightly increasing with respect to advice coverage, being relatively high, even for lower values of advice coverage. Compared to reactive coaching, proactive and reflexive coaching appear to guarantee a constant high conformity, mitigating the negative effects of arbitrarily low coverage almost equally well. In a sense, earlier coach intervention guarantees that the learner cannot deviate much from the coach's guidance, even in case where advice is rather minimal, leaving space for generalization.

6 Discussion and Future Work

In the above, a machine learning paradigm that allows for the efficacious and, under certain configurations, faithful transfer of knowledge from a coach to a machine learner has been explored. A limitation of this design is that the list sorting task is highly structured, admitting a single solution, which might not match many real-world scenarios. Nevertheless, this stiffness makes the sorting task ideal for a proof of concept study design. Moreover, the presented in depth analysis in terms of different coaching policies, coaching styles and learner memory types effectively mitigates this restriction in the context of exploring and discussing the key features of the proposed learning paradigm. Another limitation stems from the used knowledge representation, which has been artificially restricted to linearly ordered propositional knowledge bases, with rules of a task-specific form. However, utilizing richer representations, as in other works (Michael, 2019; Markos and Michael, 2022; Ioannou and Michael, 2021) would make policy description trivial, since, e.g., bubble sort can be easily described by a single such rule. This, in turn, would result to single-step coaching sessions, regardless of coaching policy, coaching style, or memory configuration, making any substantial analysis in this setting impossible and / or trivial. Hence, restricting to a highly structured yet rich problem with "expressive enough" semantics allows for an in depth initial exploration of the proposed learning paradigm while it does not deduct much value from generalizations that can be made to broader and more realistic cases.

A key outcome is that properly shaping an agent's learning environment strongly affects both the acquired knowledge and the way it is applied, allowing for a wide variety of agent behaviors, aligned with the subtleties of the actual task at hand. For instance, if just reaching a certain target state is the ultimate goal, as, e.g., in naive route finding, then using a lazy reactive coaching style allows for learning such a policy in a minimal number of steps. Furthermore, the utilization of memory by the learner appears to essentially wipe out any coaching policy differences, at least in the case explored in this work. This is a useful feature in the case of coaches with potentially imperfect policies, or when merging knowledge from multiple resources / coaches, at the cost of not having any guarantees on *how* the agent will end up solving the task.

In previous work exploring ideas similar to Machine Coaching (Ioannou and Michael, 2021; Markos et al., 2022), emphasis has been put on "what" is learned by the learner, being mostly oblivious to how the learner eventually applies any obtained knowledge to achieve its goals. In contrast, in this work ways in which "how-to" knowledge can be instilled to the learner by the coach have also been considered. Indeed, the proposed search coaching paradigm can be properly configured in plentiful ways to allow for such a more meticulous learning. As shown, one way this can be achieved is by providing overspecific advice, for virtually any coaching policy, coaching style, and / or memory configuration, essentially having the coach guide the learner step-by-step. While implying higher and more time consuming coach involvement, this kind of coaching and its independence of any other examined parameters, except for advice shape, might prove useful in the design of knowledge acquisition methodologies where there are no guarantees on the coaching policy efficiency, coaching style, or learner memory availability, e.g., when acquiring potentially conflicting knowledge from multiple semi-expert users. By utilizing proactive, or even reactive, coaching, at least in the uniform probability setting explored here, one might coach an eventually efficacious and faithful learner, with the additional advantage of utilizing just partial advice, if needed. This, again, might correspond to real-world scenarios where users are expert or more careful in the way advice is provided and the provision of full advice is impossible due to combinatorial or other restrictions.

Overall, the versatility of the search coaching paradigm presented in this work allows for a multitude of real-world applications to be considered. At the moment, work is being conducted on applying similar principles in training a cognitive agent to play Othello, utilizing expert feedback from human players. In this setting, playing Othello can be formulated as a search problem for a winning game state, while coach advice comes as actions on the basis of patterns observed on board. Furthermore, the way the learner searches for each next more separately is another search-coachable aspect of the problem, allowing for an even more fine grained control over the learner's behavior. In contrast to the controlled setting utilized in this work, more complex knowledge representation and reasoning semantics, as in (Michael, 2019; Markos and Michael, 2022) are being utilized. In the same spirit, the introduction of more complex reasoning semantics in the above setting, as the ones presented in (Dung, 1995) could be considered, aiming to address, e.g., the problem of efficiently embedding knowledge coming from multiple sources.

Given the robustness of the proposed coaching paradigm with respect to the "how-to" part of acquired knowledge application, it can also be considered in scenarios where faithfulness to human expertise and its application is crucial. In such cases, as, for example, a cognitive assistant for professional counseling, where the goal can be formulated as bringing the client to a given "target state", how this is achieved often matters most (Sedlakova and Trachsel, 2023). To that end, the presented approach guarantees that several learning configurations allow for such a faithful knowledge transfer and further guarantee the efficacy of the learned policy. Future works in that direction could consider the development and testing of cognitive assistants in such occasions, utilizing more expressive semantics aligned with the needs of the corresponding domain of application.

In summary, the search coaching paradigm demonstrates a versatile and parametrizable methodology for knowledge transfer, offering control over both the efficacy of the learned policy and its conformity to the coach's guidance. The exploration across various memory configurations and coaching styles highlights a key trade-off: a "lazy" reactive coach can enable a learner to efficiently acquire an efficacious policy, albeit with the risk of unfaithfulness, whereas proactive and reflexive coaching, or the provision of overspecific advice, guarantees a consistently conformant learner. This fine-grained ability to shape agent knowledge application makes the paradigm particularly relevant for complex, real-world search problems. Ultimately, this work provides a configurable learning framework for the development of capable and faithful cognitive representatives.

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